



Name: _____

Teacher: _____

FORT STREET HIGH SCHOOL

2012

**PRELIMINARY HIGH SCHOOL CERTIFICATE COURSE
ASSESSMENT TASK 1**

Mathematics

**TIME ALLOWED: 1 HOUR
PLUS 5 MINUTES READING TIME**

Outcomes Assessed	Questions	Marks
Demonstrates the ability to manipulate and simplify numeric and algebraic expressions	1	
Solves problems involving equations and inequalities	2	
Uses appropriate techniques to solve problems involving plane geometry	3	
Solves problems involving indices and logs	4	

Question	1	2	3	4	Total	%
Marks	/14	/14	/13	/14	/55	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each Question is to be started in a new booklet.

- a) Mr Fraser bought an antique chair valued at \$980. Each year its value appreciated by 12%. Calculate the value of the chair after 5 years. Answer correct to the nearest 5 cents.

2



- b) Calculate correct to four significant figures:

2

$$\frac{\left(\frac{2}{5}\right)^4 \times \left(\frac{3}{4}\right)^5}{\left(\frac{6}{7}\right)^2 + \left(\frac{2}{3}\right)^3}$$

- c) Expand and simplify:

2

$$\frac{12x}{x+1} \div \frac{6x}{x^2+2x+1}$$

- d) Factorise fully:

2

$$y^3 - 64$$

- e) Express the following recurring decimal as a simplest fraction.

3

$$0.\dot{2}9\dot{7}$$

- f) Express as a single fraction with rational denominator. Express your answer in simplest form.

3

$$\frac{2}{6-3\sqrt{3}} - \frac{1}{3+2\sqrt{3}}$$

Question 2 (Equations and Inequalities)14 marks

- a) Solve for a : 2

$$\frac{a}{4} - \frac{a+2}{3} = 9$$

- b) Solve for x : 2

$$\frac{x-3}{7} - \frac{3}{4} \geq 9$$

- c) Solve by completing the square. Answer in exact form. 2

$$x^2 - 10x + 20 = 0$$

- d) Solve for x : 2

$$x^2 - 11x + 18 > 0$$

- e) Solve for x : 3

$$|x-2| = 7-2x$$

- f) Solve simultaneously: 3

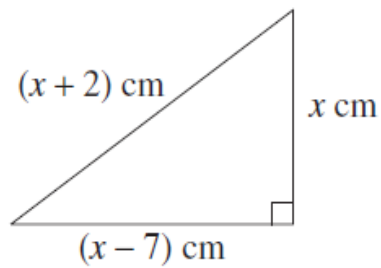
$$a+b=5$$

$$2a+b+c=4$$

$$a-b-c=5$$

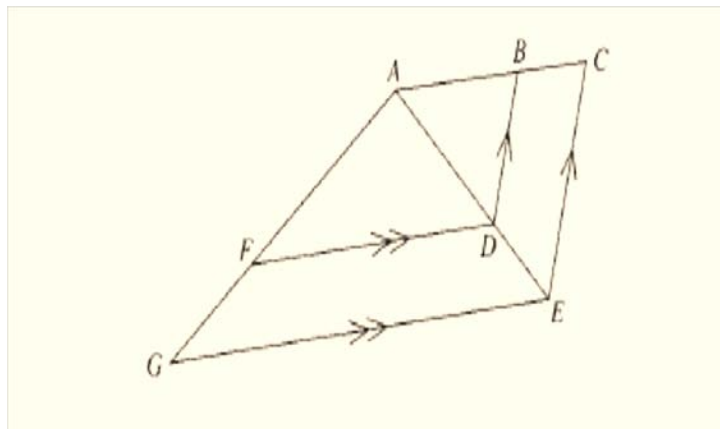
- a) The interior angles in a regular polygon are 140° . Calculate the number of sides of the polygon. 2

- b) Solve for x : 2

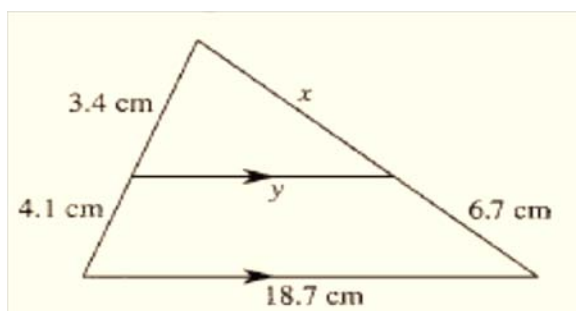


- c) Prove that $\frac{AF}{AG} = \frac{AB}{AC}$

2



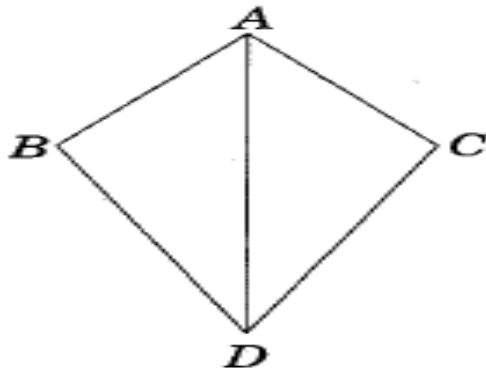
- d) Find the value of the pronumerals.
(Answer to 1 decimal place) 3



e) Given $AB = AC$ and AD bisects $\angle BAC$. Prove that 3

i) $\triangle ABD \equiv \triangle ACD$, and

ii) Hence give a reason why $DB=DC$ 1



Question 4 (Logarithms)

14 marks

a) Evaluate: 1

$$\log_6 6\sqrt{6}$$

b) Solve for x : $\log_3 x = 5$ 1

c) Given $\log_7 2 = 0.36$ and $\log_7 5 = 0.83$, find $\log_7 70$. 2

d) Solve $\log_9 200$ to 1 decimal place. 2

e) Solve: $3^{x+2} = 81$ 2

f) Evaluate:

i) $\log_5 500 - \log_5 4$ 3

ii) $2\log_3 6 + \log_3 18 - 3\log_3 2$ 3

END

2012 Assessment 1: Preliminary HSC Solutions

Question 1

(14)

a. $A = P(1+r)^n$

$$A_5 = \$980(1.12)^5 \quad \checkmark$$

(2)

$$= \underline{\underline{\$1727.10}} \quad \checkmark$$

b. $= 0.00589239 \dots \quad \checkmark$

(2)

$$= \underline{0.005892} \quad \checkmark$$

c. $\frac{12x}{x+1} \times \frac{x^2 + 2x + 1}{6x}$

$$= \frac{\cancel{12}x^2}{\cancel{x+1}} \times \frac{(x+1)(x+1)}{\cancel{6x}} \quad \checkmark \text{ factorised}$$

(2)

$$= \underline{2(x+1)} \quad \checkmark$$

d. $y^3 - 64 = y^3 - 4^3$
 $= (y - 4)(y^2 + 4y + 16)$

(2)

e) let $x = 0.297297297\dots$ — ①
 $1000x = 297.297\dots$ — ② ✓

② — ①

$$999x = 297$$

$$x = \frac{297}{999} \quad \checkmark$$

③

$$= \frac{11}{37} \quad \checkmark$$

f. $\frac{2(6+3\sqrt{3})}{(6-3\sqrt{3})(6+3\sqrt{3})} - \frac{(3-2\sqrt{3})}{(3+2\sqrt{3})(3-2\sqrt{3})}$

$$= \frac{12+6\sqrt{3}}{36-27} - \frac{(3-2\sqrt{3})}{9-12} \quad \checkmark$$

$$= \frac{12+6\sqrt{3}}{9} - \frac{(3-2\sqrt{3})3}{-3 \times 3}$$

③

$$= \frac{12+6\sqrt{3} + 9-6\sqrt{3}}{9} \quad \checkmark$$

$$= \frac{21}{9}$$

* $\frac{0}{3}$ not
 $\times$ by
conjugate

$$= \frac{7}{3} \quad (\text{or } 2\frac{1}{3})$$

✓

②

Question 2

a. $\left[\frac{a}{4} - \frac{(a+2)}{3} = 9 \right] 12$

$$3a - 4(a+2) = 108 \quad \checkmark$$

$$-a - 8 = 108$$

$$a = -116 \quad \checkmark$$

(2)

b. $\left[\frac{x-3}{7} - \frac{3}{4} \geq 9 \right] 28$

$$4(x-3) - 21 \geq 252$$

$$4x - 12 - 21 \geq 252 \quad \checkmark$$

$$4x \geq 285$$

$$x \geq \frac{285}{4} \quad \checkmark \quad \left(71 \frac{1}{4} \right)$$

(2)

c. $x^2 - 10x + (-5)^2 = -20 + 25$

$$(x-5)^2 = 5 \quad \checkmark$$

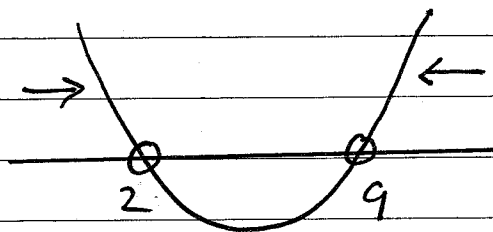
(2)

$$x-5 = \pm\sqrt{5}$$

$$x = 5 \pm \sqrt{5} \quad \checkmark$$

d. $x^2 - 11x + 18 > 0$

$$(x-9)(x-2) > 0$$



(2)

$$x < 2 \quad \text{or} \quad x > 9$$

✓

✓

(3)

e. $|x-2| = 7-2x$

$$x-2 = \pm(7-2x)$$

Case 1:

$$x-2 = 7-2x$$

$$3x = 9$$

$$x = 3$$

✓

Case 2: $x-2 = 2x-7$

$$x = +5$$

✓

check:

When $x = 3$

$$LHS = 1$$

$$RHS = 1$$

$$\therefore LHS = RHS$$

$$x = +5$$

$$LHS = 3$$

$$RHS = -3$$

$$LHS \neq RHS$$

(3)

$\therefore x = 3$ is the only solution. ✓

f. $a+b = 5$ ——— (1)

$$2a+b+c = 4$$
 ——— (2)

$$a-b-c = 5$$
 ——— (3)

$$(2) + (3)$$

$$3a = 9$$

$$a = 3$$

✓

Sub $a = 3$ into (1)

$$b = 2$$

✓

(3)

Sub $a = 3, b = 2$ into 5

$$3-2-c = 5$$

$$c = -4$$

✓

(4)

Question 3

a. Interior angle $= 140^\circ$

$$\therefore \text{Exterior angle} = 40^\circ$$

$$\text{Sum of exterior angles} = 360^\circ$$

$$\therefore \text{number of sides} = \frac{360^\circ}{40^\circ} \quad \checkmark$$

$$= 9 \quad \checkmark \quad (2)$$

b. $(x+2)^2 = x^2 + (x-7)^2$

$$\cancel{x^2} + 4x + 4 = \cancel{x^2} + x^2 - 14x + 49$$

$$x^2 - 18x + 45 = 0 \quad \checkmark$$

(2)

$$(x-3)(x-15) = 0$$

$$x = 3 \text{ or } 15 \quad \checkmark$$

c) $\frac{AF}{AG} = \frac{AD}{AE} \quad \checkmark$

—— (1) ($\triangle AGE$)
Ratio properties of a triangle

Also $\frac{AD}{AE} = \frac{AB}{AC} \quad \checkmark$ —— (2) ($\triangle ACE$)

Sub (1) into (2)

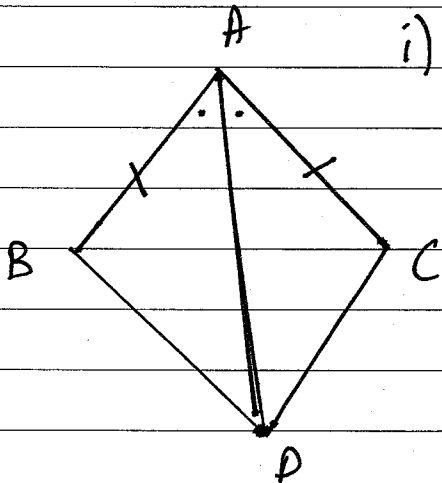
(2)

$$\therefore \frac{AF}{AG} = \frac{AB}{AC}$$

(5)

d.	$\frac{x}{3.4} = \frac{6.7}{4.1}$ $x = 5.6 \text{ cm}$	$\frac{x}{y} = \frac{(x+6.7)}{18.7}$ $y = \frac{5.6(18.7)}{12.3}$ $= 8.5 \text{ cm}$
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e.



i) In $\triangle ABD$ and $\triangle ACD$

- $AB = AC$ (given) ✓
- $\angle BAD = \angle CAD$ (given) ✓
- AD is common ✓

$\therefore \triangle ABD \equiv \triangle ACD$ (SAS) ✓

ii) $\therefore DB = DC$ (Corresponding sides of congruent triangles) ✓

Question 4

a) $\log_6 6\sqrt{6} = \log_6 6^{\frac{3}{2}}$

$$= \frac{3}{2} \log_6 6$$

$$= \frac{3}{2} \quad \checkmark \text{ (with working) shown} \quad (1)$$

b) $\log_3 x = 5$

$$x = 3^5$$

$$= 243 \quad \checkmark$$

(1)

c) $\log_7 70 = \log_7 (7 \times 5 \times 2)$

$$= \log_7 7 + \log_7 5 + \log_7 2 \quad \checkmark$$

$$= 1 + 0.83 + 0.36$$

$$= 2.19 \quad \checkmark$$

(2)

d) $\log_9 200 = x$

$$9^x = 200$$

$$x = \frac{\ln 200}{\ln 9} \quad \checkmark$$

$$= 2.4 \quad \checkmark$$

(2)

(7)

e. $3^{x+2} = 81$ or $x+2 = \frac{\ln 81}{\ln 3}$

$$3^{x+2} = 3^4 \quad \checkmark$$

$$x+2 = 4$$

$$x = 2 \quad \checkmark$$

(2)

$$x+2 = 4$$

$$x = 2$$

f.

i) $\log_5 500 - \log_5 4$

$$= \log_5 \left(\frac{500}{4} \right) \quad \checkmark$$

$$= \log_5 125 \quad \checkmark$$

$$= \log_5 5^3$$

$$= 3 \log_5 5$$

$$= 3 \quad \checkmark$$

or

$$= \frac{\ln 125}{\ln 5}$$

$$= 3$$

(3)

ii) $\log_3 6^2 + \log_3 18 - \log_3 2^3$

$$= \log_3 \left(\frac{36 \times 18}{8} \right) \quad \checkmark$$

$$= \log_3 81 \quad \checkmark$$

$$= \log_3 3^4$$

$$= 4 \log_3 3$$

$$= 4 \quad \checkmark$$

(3)

END

(8)